

## MATERIAL SAFETY DATA SHEET

### 1) Identification of Substance and Company

**Trade Name:** DOP

**Chemical Name:** Di(2-ethylhexyl) Phthalate

**Product Use:** DOP is used as a Plasticizer in PVC resins for fabricating flexible vinyl Product like.

**Manufacturer Name / Supplier:** South City Petrochem Co, Ltd.

**Office:** 258 Soi Pattanakarn 30, Pattanakarn Rd.,

Suanluang, Suanluang, Bangkok 10250

Tel. 66 (0) 27171730-48

Fax.66 (0) 27171750

**Emergency Call:** 038-968546-9

**Homepage :** [www.southcitygroup.com](http://www.southcitygroup.com)

**Factory:** 99/1 Moo 8, Main Road No 3191.

Tambol Mab Kha Branch District.

Nikom Pattana, Rayong 21180

Tel: 038 968546-9 Fax 038 636 301

### 2. Hazard Identification

#### 2.1) GHS classification

**Physical Hazards:** -

#### Health Hazards

**Skin Corrosion/Irritation** : Category 3

**Serious Eye Damage/Eye Irritation :** Category 2B

**Reproductive Toxicity** : Category 1B

#### Environmental Hazards

**Chronic Aquatic Toxicity** : Category 4

## 2.2) GHS-Labeling Regulation

### 2.2.1) Pictograms



Signal words : danger

May damage fertility or the unborn child.

### 2.2.2) Hazardous statements

#### Health Hazards

Causes mild skin irritation.

Causes eye irritation.

May damage fertility or the unborn child.

#### Environmental Hazards

Harmful to aquatic life with long lasting effects.

### 2.2.3) Precautionary statements and pictograms

Use with local exhaust ventilation. Avoid all sources of ignition: heat, sparks, open flame.

Materials and containers of material. Must be disposed of as hazardous waste.

Absorb the spill with sand or inert absorbent materials and keep in a safe.

Wear appropriate personal protective equipment safety.

Avoid direct contact. If you touch or feel unwell should seek medical attention immediately.

## 3. Composition/Information on Ingredients

### Substance / Mixture :

| Chemical Name                     | CAS NO.  | Conc.         |
|-----------------------------------|----------|---------------|
| Di(2-ethylhexyl) Phthalate        | 117-81-7 | $\geq 99.5\%$ |
| 2-Ethylhexanol                    | 104-76-7 | $< 0.05\%$    |
| Other (Water and Volatile Matter) | -        | $< 0.45\%$    |

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## 4. First-aid measures

|                      |                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>If in eyes:</b>   | Wash affected eyes for at least 15 minutes under running water with eyelids held open.<br>Seek medical attention.                    |
| <b>If on skin:</b>   | Remove contaminated clothing. Wash affected areas thoroughly with soap and water.<br>If irritation develops, seek medical attention. |
| <b>If inhaled:</b>   | Keep patient calm, remove to fresh air. If breathing difficulties develop, aid in breathing and seek immediate medical attention.    |
| <b>If swallowed:</b> | Rinse mouth immediately and then drink plenty of water, seek medical attention.                                                      |

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## 5. Fire-fighting measures

**Substances to avoid:** -

**Protection against fire and explosion:** Avoid all sources of ignition: heat, sparks, open flame.

**Suitable extinguishing media:** carbon dioxide, dry powder, foam

**Potential hazards :** Vapour are heavier than air and may spread along floors. Forms explosive mixtures with air at elevated temperatures. Development of hazardous combustion gases or vapour possible in the event of fire.

**Protective equipment for fire-fighting:** approved self-contained breathing apparatus should be worn.

Use water spray to cool fire-exposed containers.

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## 6. Accidental release measures

**6.1 Personal precautions:** Handle in accordance with good industrial hygiene and safety practice.

**6.2 Environmental precautions:** Do not empty into drains. Avoid contaminating for the environment.

**6.3 Methods for cleaning up or tanking up:**

For large amounts: Pump off product.

For residues: Pick up with suitable adsorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulation.

## 7. Handling and storage

### 7.1 Handling

Ensure thorough ventilation of stores and work areas.

Wear appropriate personal protective equipment safety.

Avoid contact with the skin, eyes and clothing.

### 7.2 Storage

Keep container tightly closed.

Avoid all sources of ignition: heat, sparks, open flame.

### 7.3 Storage incompatibility:

None

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## 8. Exposure controls /personal protection

### 8.1 The average concentration in the atmosphere has been established to work:

TLV : 5 mg/m<sup>3</sup>

TWA : 5 mg/m<sup>3</sup>

### 8.2 Engineering measures :

Technical measures and appropriate working operations should be given priority over the use of personal protection equipment.

### 8.3 Personal protective equipment

#### Respiratory protection:

Wear respiratory protection if ventilation is inadequate. Wear a NIOSH-certified (or equivalent) organic vapour/particulate respirator as needed.

#### Eye protection:

Safety glasses .

#### Hand protection:

Nitrile Gloves.

#### Body protection:

Body protection must be chosen based on level of activity and exposure.

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### 8.3 General safety and hygiene measures:

Change contaminated clothing. Wash hands and face after working.

Avoid inhalation of mists. Avoid contact with the skin, eyes and clothing.

No smoking.

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## 9. Physical and chemical properties

**Form:** Liquid

**Color :** Colorless

**Oder :** characteristic odor

**Boiling Point:** 385°C @760 mmHg

**Melting Point:** -50 °C

**Freezing Point:** -50°C

**Vapor Density:** 13.5

**pH:** Neutral @ 20°C

**Specific Gravity:** 0.986 g/cm<sup>3</sup> @ 20 °C

**Viscosity :** 70-80 mPa.s @ 20 °C

**Vapor Pressure :** 0.001 kPa @ 20 °C

**Flash Point :** 215°C (Closed-cup Test)

**Solubility in Water:** 0.1 g/l @ 20 °C

**Explosion limits** lower < 0.1 Vol%

**Solubility in Organic Solvent:** Soluble

Upper 0.2 Vol%

**Autoignition temperature:** N/A

**Ignition temperature:** ≈ 350°C

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## 10. Stability and reactive

**10.1 Reaction :** Vapour / air-mixture are explosive at intense warming.

**10.2 Stability :** This product is stable under normal conditions.

**10.3 Hazardous reactions may occur:** Severe reactions to substances that cause oxidation.

**10.4 Conditions to avoid:** Avoid all sources of ignition: heat, sparks, open flame.

**10.5 Substances to avoid:** None

**10.6 Toxic substances / A Hazardous decomposition products:** None

## 11. Toxicological information

**Acute Oral toxicity** : LD50 (mouse) >30,600 mg/kg

**Symptoms** : Nausea, Vomiting, Mucosal irritation, windpipe, esophagus and Digestive system.

**Acute Inhalation toxicity** : LC50 (Rat, 4 h) >10.6 mg/l

**Symptoms** : Mucosal irritation.

**Acute Dermal toxicity** : LD50 (Rabbit) >25,000 mg/kg

**Skin Irritation / corrosion** (rabbit) : Slightly irritating.

**Serious Eye Damage/Eye Irritation** (rabbit) : Slightly irritating.

**Sensitisation** : Sensitisation test

Result : Negative

**Genetic toxicity** : Ames-test

Result : Negative

### CMR

The animal study found. Chemical trend to cause abnormalities in the fetus.

Experiments have shown reproductive toxicity in animal studies, female and male.

It is assumed that affect the human reproductive system.

## 12. Ecological information

### 12.1 Toxicity

**Acute:** Fish : LC50 (48 h) >1,000 mg/L ; leuciscus idus

Bacteria : EC10 (6 h) >1,000 mg/L ; Pseudomonas putid

### 12.2 Degradability / Persistence

**Biological / Abiological Degradation** : 62% ; 28 day

**Test method OECD 301C** : Biodegradation can be difficult.

### 12.3 Bioaccumulation

**Partitioning coefficient n-octanol/water (log Pow)** : 7.60

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### 12.4 Mobility in Soil

No information available.

### 12.5 Result of PBT and vPvB assessment

PBT/vPvB assessment not available as chemical safety assessment not required/not conducted.

### 12.6 Other adverse effects

Avoid release to the environment.

## 13. Disposal considerations

### 13.1 chemical substance disposal :

Keep the spill in a closed container for disposal.

Absorb the spill with sand or inert absorbent materials and keep in a safe.

Wear appropriate personal protective equipment .

### 13.2 Container disposal:

Materials and containers of material. Must be disposed of as hazardous waste.

### 13.3 Waste disposal of substance:

Dispose of in a licensed facility. Do not discharge substance/product into sewer system. Dispose of in accordance with national, state and local regulations.

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## 14. Transport information

### Land transport

#### USDOT

DOT (USA) :

DOT Classification :- Not a DOT controlled material (United State)

Class 9, UN 3082, Packing Group III

when material is shipped in quantities in one package at or above the Reportable

Quantity and when no other hazard class applies; not regulated.

Possible Shipping Description(s) : 9 UN 3082 III

Environmentally hazardous substance, liquid, n.o.s

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## Sea transport

### Sea-IMG (International Maritime Dangerous Goods)

Shipping Information :

HS Code : 2917 34 00

USCG Compatability Group : 34 Esters (Oxidizing agents such as sulfuric acid and Nitric acid)

USCG CHRIS Code : EHE

Storage Temperature : Ambient

Inert atmosphere : No requirement

Venting : Open

IMDG Code : 9 UN 3082 Packing group III

Possible Shipping Description(s): Environmentally hazardous substance, liquid, n.o.s.

## Air transport

### Air - ICAO (International Civil Aviation Organization)

Class not regulated

Possible Shipping Description(s):

Not regulated

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## 15. Regulatory Information

### 15.1 Safety health and environmental regulations/legislation specific of the substance or mixture

Category Management : 6.1A

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## 16. Other information

**Reference** 1. <http://www.plassicisers.org/>

2. <http://ecpi.org/>